30 MCQ Questions: Database Fundamentals

Comprehensive Single-Page Quick Review Sheet

1. What is the primary purpose of a DBMS?
A) Store & manage data
B) Speed up hardware
C) Compile code
D) Design web pages
Answer: A) Store & manage data efficiently

2. Raw, unprocessed facts are defined as:
A) Information
B) Data
C) Knowledge
D) Metadata
Answer: B) Data

3. Processed data that provides meaningful context is:
A) Metadata
B) Information
C) Database
D) Program
Answer: B) Information

4. The term "Metadata" is best defined as:
A) Big Data
B) Data about Data
C) Deleted Data
D) Duplicate Data
Answer: B) Data about Data

5. A major flaw of traditional file processing systems is:
A) Data Redundancy
B) High Security
C) Easy Sharing
D) Low Storage Cost
Answer: A) Data Redundancy

6. The most widely used data model in modern DBMS is:
A) Hierarchical
B) Relational Model
C) Network Model
D) Flat File
Answer: B) Relational Model

7. In relational database theory, a table is called a:
A) Relation
B) Tuple
C) Attribute
D) Domain
Answer: A) Relation

8. A single row in a relational table is formally called a:
A) Attribute
B) Tuple
C) Schema
D) Domain
Answer: B) Tuple

9. A column in a relational table represents an:
A) Attribute
B) Tuple
C) Relation
D) Instance
Answer: A) Attribute

10. The overall logical structure and design of a DB is:
A) Schema
B) Instance
C) State
D) Query
Answer: A) Schema

11. The actual content stored in a DB at a specific moment:
A) DB Instance
B) DB Schema
C) Metadata
D) Architecture
Answer: A) DB Instance

12. In ER Diagrams, an Entity is represented by a:
A) Oval
B) Rectangle
C) Diamond
D) Line
Answer: B) Rectangle

13. In ER Diagrams, an Attribute is represented by an:
A) Oval / Ellipse
B) Rectangle
C) Diamond
D) Triangle
Answer: A) Oval / Ellipse

14. In ER Diagrams, a Relationship is represented by a:
A) Circle
B) Diamond
C) Square
D) Dotted Line
Answer: B) Diamond

15. A Weak Entity is an entity that:
A) Lacks a Primary Key
B) Has no attributes
C) Is not in a table
D) Contains duplicate data
Answer: A) Lacks a Primary Key on its own

16. An attribute that can be divided into smaller parts:
A) Composite Attribute
B) Derived Attribute
C) Multi-valued
D) Simple Attribute
Answer: A) Composite Attribute

17. An attribute holding multiple values for a single entity:
A) Single-valued
B) Multi-valued
C) Derived
D) Stored
Answer: B) Multi-valued Attribute

18. The "Degree" of a relationship refers to the:
A) Number of entities involved
B) Number of table rows
C) Number of columns
D) Size of the database
Answer: A) Number of participating entities

19. A unique identifier for a row that cannot be NULL:
A) Foreign Key
B) Primary Key
C) Alternate Key
D) Composite Key
Answer: B) Primary Key

20. A key in a table that references a Primary Key in another:
A) Primary Key
B) Foreign Key
C) Super Key
D) Candidate Key
Answer: B) Foreign Key

21. A key composed of two or more attributes together:
A) Composite Key
B) Foreign Key
C) Secondary Key
D) Unique Key
Answer: A) Composite Key

22. Any set of attributes that uniquely identifies a row:
A) Super Key
B) Foreign Key
C) Secondary Key
D) Index
Answer: A) Super Key

23. Candidate Keys are defined as:
A) Minimal Super Keys
B) All Foreign Keys
C) Duplicate Keys
D) Secondary Indices
Answer: A) Minimal Super Keys

24. Entity Integrity Constraint enforces that:
A) Primary key cannot be NULL
B) FK must match PK
C) Data must be numbers
D) No keys are allowed
Answer: A) Primary Key cannot be NULL

25. Referential Integrity Constraint ensures that:
A) Foreign Key values match PK
B) Table names are unique
C) Columns are non-empty
D) Users have passwords
Answer: A) Foreign Key values must match PK or be NULL

26. The set of allowable values for an attribute is called:
A) Domain
B) Tuple
C) Relation
D) Schema
Answer: A) Domain

27. Logical Data Independence allows changing the:
A) Conceptual schema without ap...
B) Hard drive location
C) Hardware server
D) User passwords
Answer: A) Conceptual schema without affecting apps

28. Physical Data Independence allows modifying:
A) Physical storage without logical ...
B) App code
C) Column names
D) Query syntax
Answer: A) Physical storage structures independently

29. The ANSI-SPARC Three-Schema Architecture consists of:
A) Internal, Conceptual, External
B) Primary, Secondary, Tertiary
C) Input, Process, Output
D) Data, Metadata, Log
Answer: A) Internal, Conceptual, External levels

30. A Data Dictionary (Data Repository) stores:
A) User passwords only
B) Database schema & definitions
C) Raw data values
D) External file backups
Answer: B) Database schema, definitions & constraints